

LAB ASSIGNMENT 8

PROBLEM 1: Apply K-Means Algorithm of all three given datasets and check their performance.

CODE:

```
import numpy as np
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, silhouette_score
from scipy.stats import mode

df_d = pd.read_csv('diabetes_dataset.csv')
df_s = pd.read_csv('Social_Network_Ads.csv')
df_t = pd.read_csv('titanic.csv')

df_d.fillna(df_d.mean(numeric_only=True), inplace=True)
df_s.fillna(df_s.mean(numeric_only=True), inplace=True)

df_t['Age'] = df_t['Age'].fillna(df_t['Age'].median())
df_t['Embarked'] =
df_t['Embarked'].fillna(df_t['Embarked'].mode()[0])
df_t = df_t.drop(columns=['Cabin'])

datasets = {
    "Diabetes Dataset": df_d,
    "Social Dataset": df_s,
    "Titanic Dataset": df_t
}

for name, df in datasets.items():
    print(f"\n{name}")
    if name == "Diabetes Dataset":
        y_true = df['Outcome'].values
        X = df.drop(columns=['Outcome'])
    elif name == "Social Dataset":
        y_true = df['Purchased'].values
        X = df.drop(columns=['Purchased'])
```

```
elif name == "Titanic Dataset":
    y_true = df['Survived'].values
    X = df.drop(columns=['Survived'])

X = X.select_dtypes(include=[np.number]).values
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
labels = kmeans.fit_predict(X_scaled)

print("\nCluster Centers:\n", kmeans.cluster_centers_)
print("\nCluster Counts:")
print(pd.Series(labels).value_counts())
print("\nInertia :", kmeans.inertia_)
sil_score = silhouette_score(X_scaled, labels)
print("Silhouette Score:", sil_score)

labels_reshaped = labels.reshape(-1, 1)
mapped_labels = np.zeros_like(labels_reshaped)
for i in range(2):
    mask = (labels_reshaped == i)
    if np.sum(mask) == 0:
        continue
    mapped_labels[mask] = mode(y_true[mask.flatten()],
keepdims=True)[0]
acc = accuracy_score(y_true, mapped_labels)
print("Accuracy:", round(acc, 4))
```

OUTPUT:

```

PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_10_temp.py"

Diabetes Dataset

Cluster Centers:
[[ 0.96221087  0.42871022  0.41023571 -0.15260198 -0.03810374  0.11897688
   0.02766651  1.05854237]
 [-0.52168059 -0.23243325 -0.22241695  0.08273601  0.02065866 -0.06450554
  -0.01499991 -0.57390851]]

Cluster Counts:
1    498
0    270
Name: count, dtype: int64

Inertia : 5128.807215743139
Silhouette Score: 0.19603986813414287
Accuracy: 0.6745

Social Dataset

Cluster Centers:
[[-0.79905703 -0.09926073 -0.31964695]
 [ 0.82339379  0.1022839  0.32938239]]

Cluster Counts:
0    203
1    197
Name: count, dtype: int64

Inertia : 890.6490213549357
Silhouette Score: 0.2506390925883895
Accuracy: 0.6425

Titanic Dataset

Cluster Centers:
[[-0.0364489  0.54689082 -0.28334331  0.04008589  0.01702918 -0.34148135]
 [ 0.09293744 -1.39446264  0.72246899 -0.10221105 -0.04342101  0.87070941]]

Cluster Counts:
0    640
1    251
Name: count, dtype: int64

Inertia : 4211.863294509498
Silhouette Score: 0.29334888624037486
Accuracy: 0.6667
PS D:\NITJ ACADEMICS\OPENLEARN>

```